# FoundationStereo: Zero-Shot Stereo Matching

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**Presented by:** Danil Smirnov, Federico Cognolatto • 3D Data Processing Poster Session • University of Padova, 2026

## Motivation

Many deep stereo matching models still rely on **per-domain fine-tuning** to achieve competitive performance.

- **Sim-to-real gap:** Synthetic training fails on real scenes
- **Challenging structures:** Reflections, textureless, thin objects
- **Weak zero-shot:** Poor generalization to unseen domains

## Key Contributions

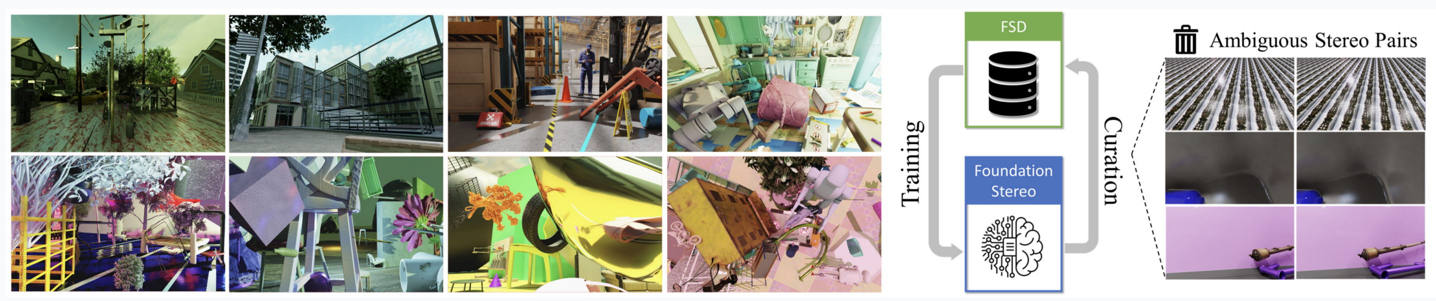
1. **FoundationStereo:** Stereo foundation model with strong zero-shot generalization
2. **FSD Dataset:** 1M synthetic stereo pairs with high diversity
3. **Side-Tuning Adapter:** Adapts DepthAnythingV2 for stereo matching
4. **Attentive Hybrid Cost Filtering:** Local + global reasoning

## FSD Dataset

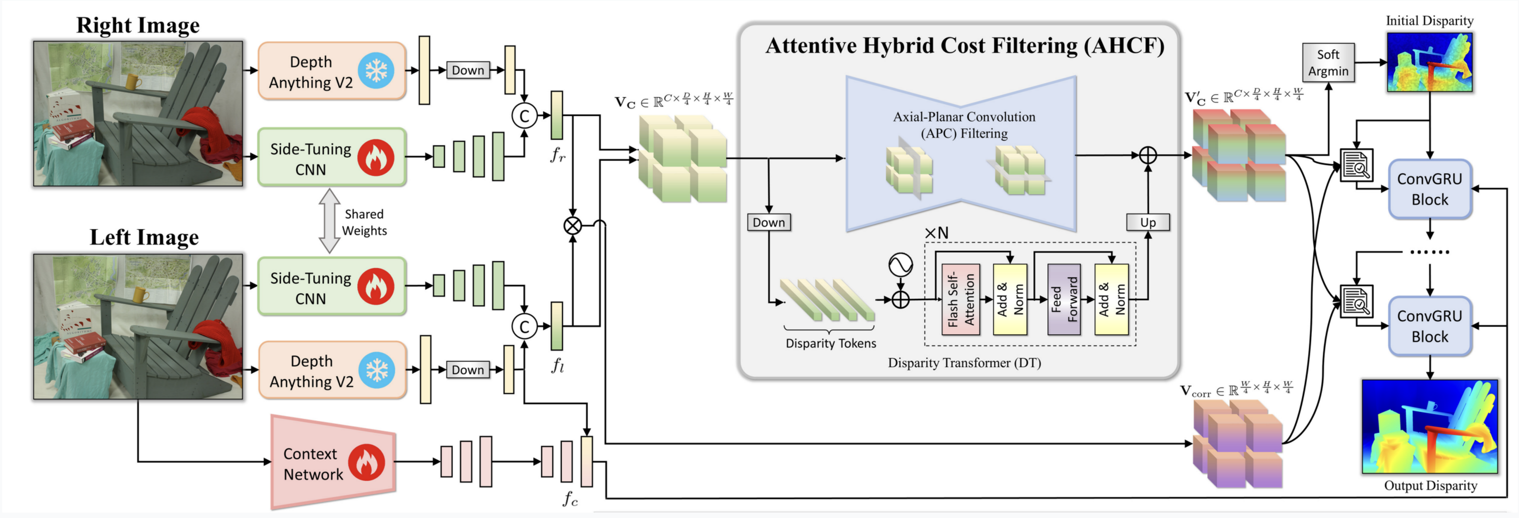
**1 million stereo pairs** rendered via NVIDIA Omniverse with high realism.

| 1280×720 | 1M |
| --- | --- |
| Resolution | Stereo Pairs |

- **Scenes:** Indoor, outdoor, driving, flying
- **Diversity:** Baselines, intrinsics, lighting
- **Self-curation:** Removes ambiguous synthetic stereo pairs




## Architecture Overview




| Step 1 | Step 2 | Step 3 | Step 4 |
| --- | --- | --- | --- |
| Feature Extraction | Hybrid Cost Volume | Cost Filtering | GRU Refinement |

## Zero-Shot Results

**State-of-the-art zero-shot performance** without target-domain fine-tuning.

| Dataset | Metric ↓ | FoundationStereo |
| --- | --- | --- |
| Middlebury | BP-2 | **1.1** |
| ETH3D | BP-1 | **0.5** |
| KITTI-12 | D1 | **2.3** |
| KITTI-15 | D1 | **2.8** |

Lower is better. Values are error rates (%).

## Key Ideas

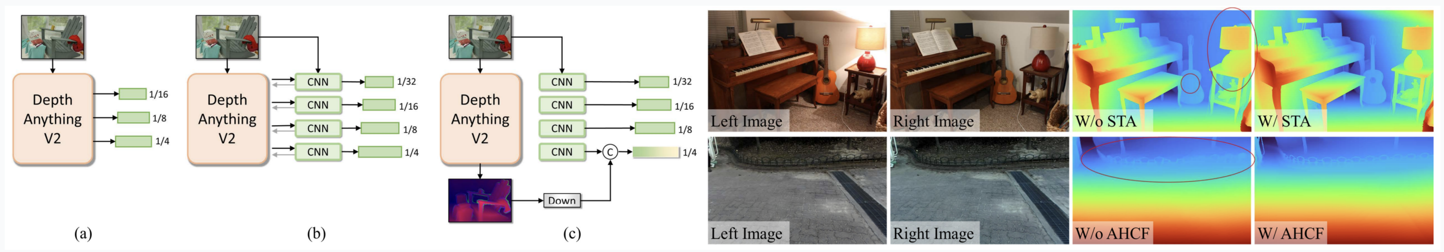
**Side-Tuning Adapter (STA)**

Adapts monocular depth features from **DepthAnythingV2** for stereo matching through lightweight adapters.

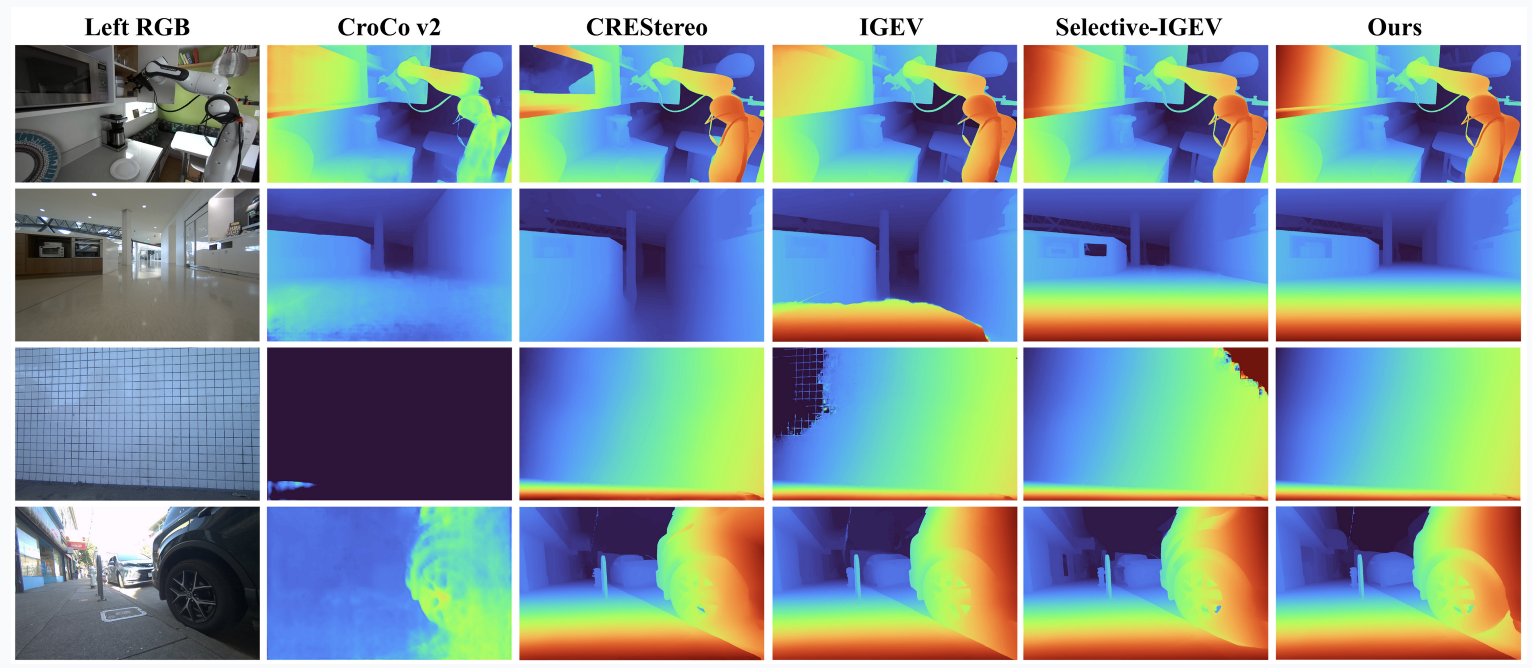
**Attentive Hybrid Cost Filtering**

→ **Axial-Planar Conv:** Efficient spatial + disparity cost-volume filtering

→ **Disparity Transformer:** Long-range reasoning over disparity space




## In-the-Wild Results




Sharp, detailed disparity maps with accurate boundaries. Better handling of reflections, thin structures, and textureless regions.

## Key Takeaways

✓ **Foundation model paradigm** works for stereo

✓ **Large-scale synthetic data** + monocular priors enable zero-shot transfer

✓ **Hybrid reasoning** crucial for robust cost volumes

## Limitations

- **Efficiency:** Transformer modules need optimization for real-time
- **Transparent objects:** Limited glass/water diversity in training